IBM HR Analytics for Employee Attrition

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ABSTRACT

Employee attrition has become a growing concern for many organizations, particularly within the technology and corporate sectors. The voluntary or involuntary departure of skilled employees can significantly disrupt organizational stability, productivity, and long-term growth. High attrition rates often lead to increased recruitment and training costs, loss of institutional knowledge, and reduced team morale. Therefore, it is essential for organizations to systematically analyze the underlying causes of employee turnover and develop strategic interventions to improve retention. This study utilizes the IBM HR Analytics Employee Attrition and Performance dataset to examine key factors influencing employee resignation. Data preprocessing techniques were applied, including the removal of missing values to ensure accurate analysis. Statistical methods such as ANOVA and Chi-Square tests were conducted to identify significant relationships between employee attributes and attrition patterns.

INTRODUCTION

In today's competitive business environment, organizations across India and globally face substantial challenges in attracting and retaining talented professionals. Employee attrition, whether voluntary or due to organizational restructuring, presents serious operational and financial concerns. The departure of experienced employees not only affects short-term performance but also disrupts workflow efficiency, team cohesion, and organizational culture. Over time, persistent turnover can weaken a company's competitive advantage and reputation in the industry.

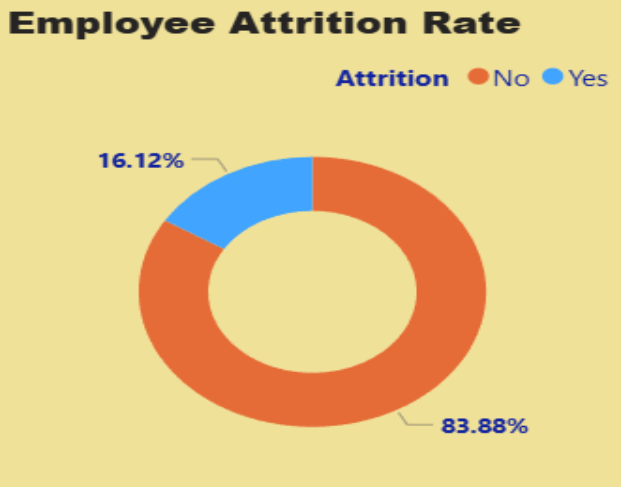
Employees represent a critical asset to any organization, as their skills, commitment, and engagement directly influence productivity and overall success. This study aims to examine employee behavior and identify the primary factors contributing to job dissatisfaction and turnover intentions. By analyzing patterns within employee demographics, job roles, satisfaction levels, and performance indicators, the research seeks to uncover meaningful insights into attrition trends.

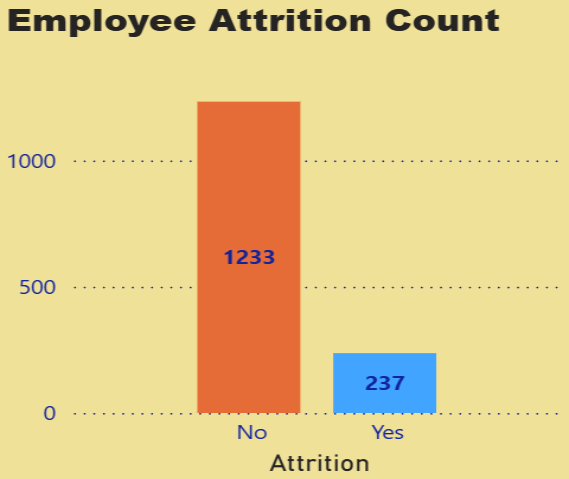
Understanding these factors enables management to implement targeted retention strategies, improve workplace policies, and foster a supportive work environment. Proactively addressing the drivers of attrition is essential for maintaining organizational stability, enhancing employee satisfaction, and ensuring sustainable long-term growth.

DATA VISUALIZATION

By analyzing employee data, we can identify factors that contribute to employee attrition, such as job satisfaction, compensation, and work-life balance. This information can be used to develop strategies to retain top talent and reduce turnover rates. HR analytics can help identify high- performing employees by analyzing data related to performance metrics, such as productivity, quality, and customer satisfaction. This information can be used to develop strategies to retain top talent and improve overall organizational performance.

1]Visualizing Employee Attrition Rate

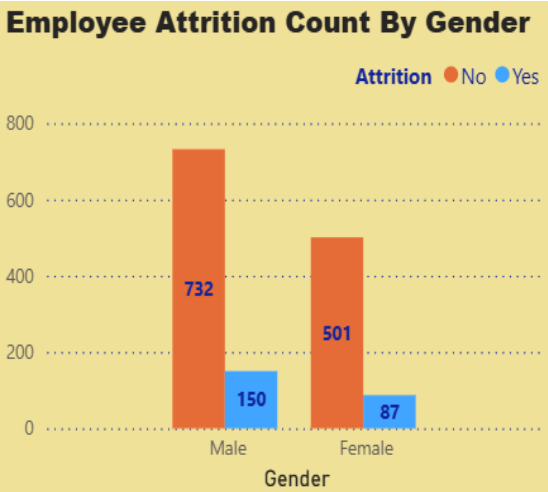


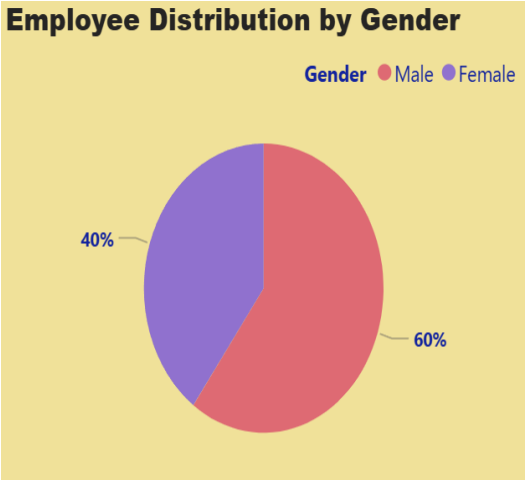


Inference:

- The employee attrition rate of this organization is 16.12%.
- So, we can say the attrition rate of the organization is at a dangerous level.
- Therefore, the organization should take measures to reduce the attrition rate.

2]Visualizing Employee Attrition by Gender

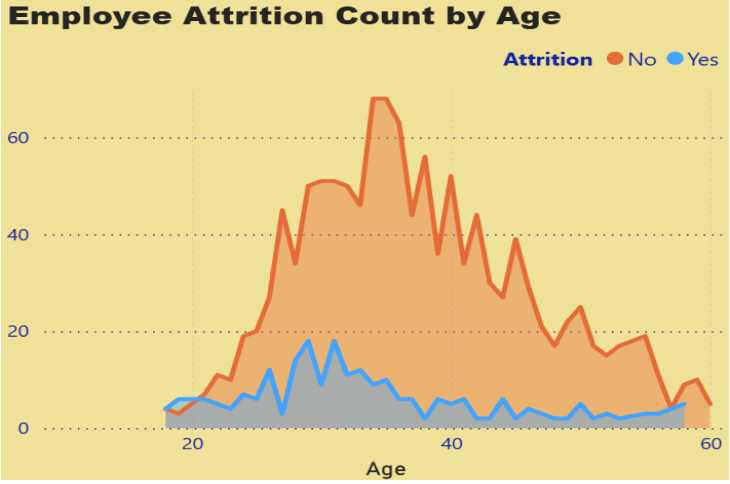




Inference:

- The number of male employees in the organization accounts for a higher count than female employee.
- Male employees are leaving more from the organization compared to female employees.

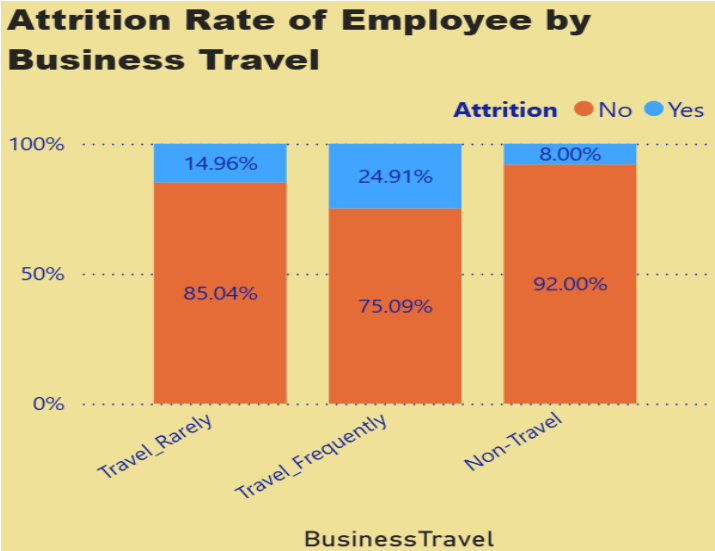
3]Employee Attrition Count by Age



Inference:

- The employee attrition count is higher in younger employees than in older employees.
- Hence, we can see that younger employees are more likely for attrition than older employees.

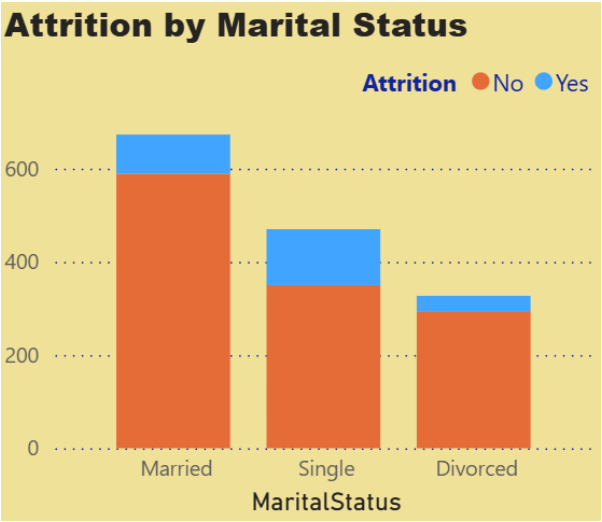
4]Employee Attrition Rate by Business Travel



Inference:

- The employees that travel more frequently have more attrition rate than employees who travel rarely and those who do not travel.
- Employees that travel more are more likely to leave the company

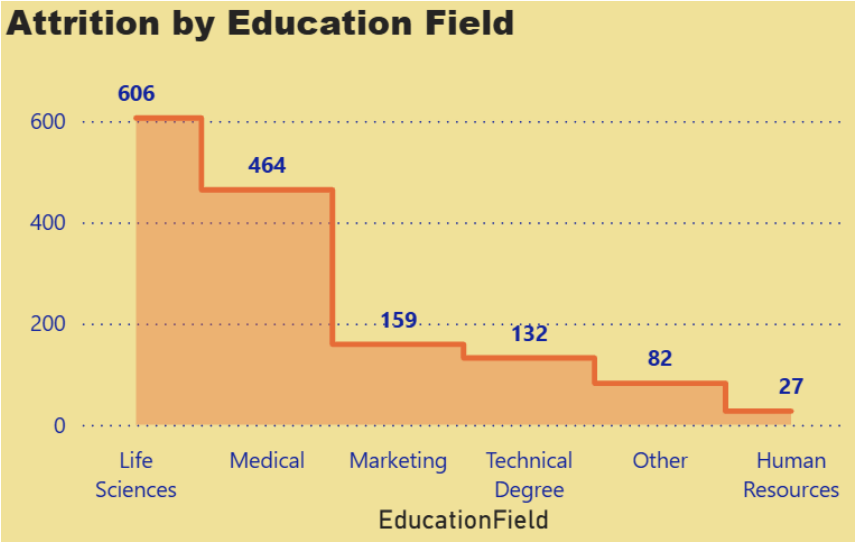
5]Employee Attrition by Marital Status



Inference:

- Divorced and married employees have very low attrition count
- Meanwhile, Single employees have high attrition count

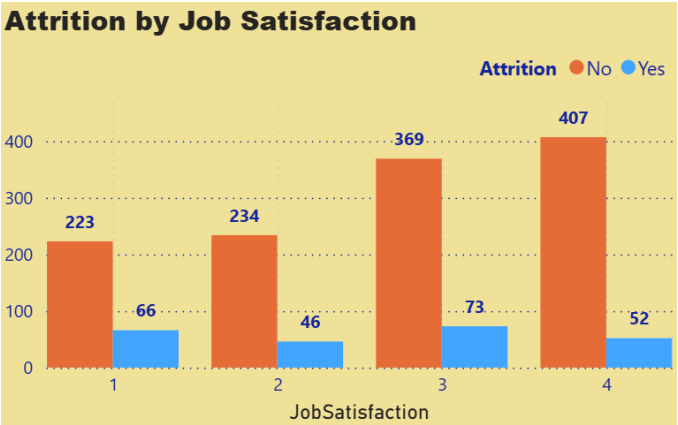
6]Employee Attrition by Education Field



Inference:

- We can see the attrition rate according to the Education Field is divided into two parts
- Employees from Life Sciences and Medical fields have high attrition rates and remaining fields have lower attrition rates

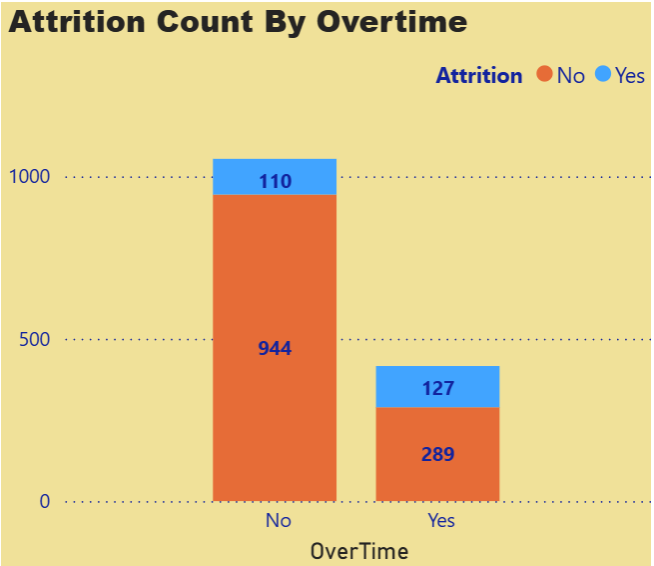
7]Employee Attrition by Job Satisfaction



Inference:

- Employees with lower job satisfaction have high attrition count and ratio.
- Meanwhile, employees with higher job satisfaction have very low attrition ratio.
- We can conclude that attrition ratio goes lower as the job satisfaction increases.

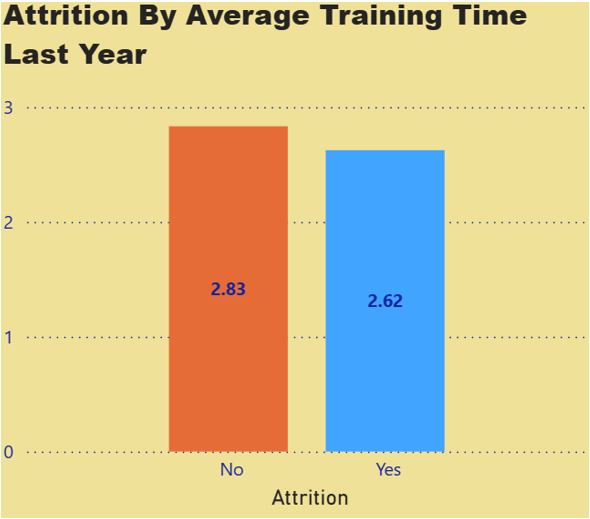
8]Employee Attrition by Overtime



Inference:

- Employees that do not overtime have higher count that of those who do, but employees doing overtime have more attrition ratio than who do not.
- We can state that overtime is one of the reasons for attrition count

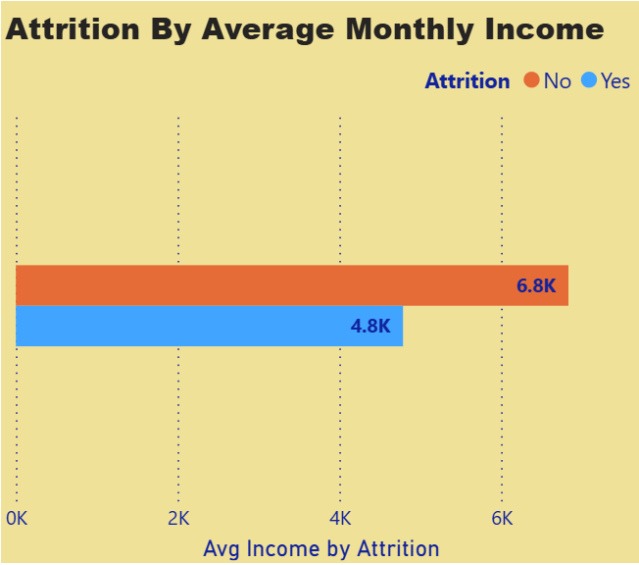
9]Employee Attrition by Training Time



Inference:

- Both the employees which left and which stayed have around the same average training time.
- Those who left had slightly lower average training time.

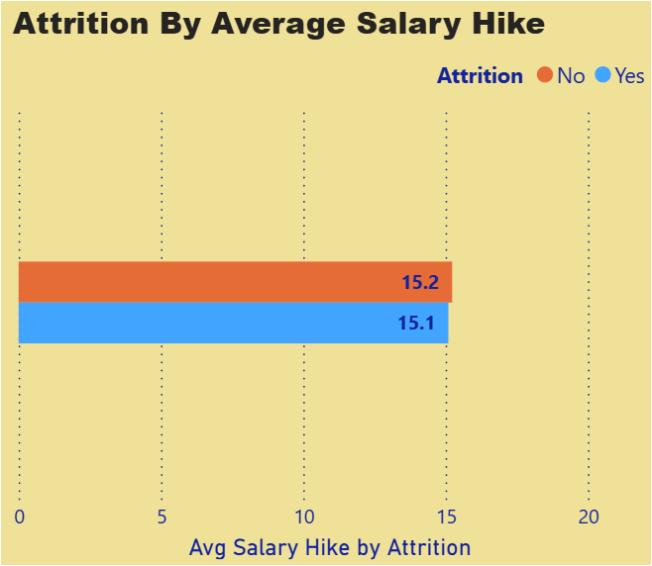
10]Employee Attrition by Monthly Income



Inference:

- Employees that left the company had lower average monthly salary than those who stayed.
- Low salary is one of the reasons for attrition

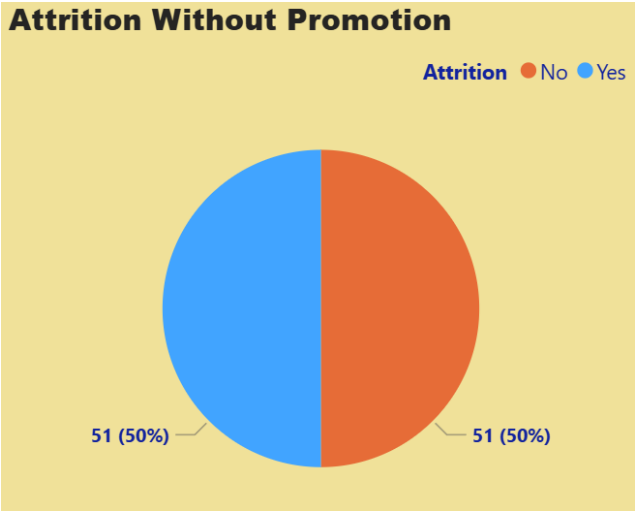
11]Employee Attrition by Salary Hike



Inference:

- Average salary hike is almost same for both the cases.
- Salary hike is not causing the high attrition rate.

12]Employee Attrition by Promotion



Inference:

- Among the employees which did not get any promotion, half of those left.
- This is one of the reasons behind high attrition rate in the company

Conclusion:

The analysis shows that the overall attrition rate is **16.12%**. Employee turnover is higher among younger and single employees. Attrition is also strongly associated with frequent business travel, lower job satisfaction levels, lower average monthly income, and lack of salary increments.

These patterns indicate that compensation, satisfaction, and work-life balance factors are key variables influencing employee attrition in the dataset. The dashboard effectively highlights how demographic and job-related factors correlate with turnover trends.